

FINAL PROJECT

FINAL PROJECT TITLE (ENGLISH)



By:

FULL NAME

XX/XXXXXX/SV/XXXXX

APPLIED BACHELOR'S DEGREE PROGRAM
ELECTRICAL ENGINEERING TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING AND INFORMATICS
VOCATIONAL COLLEGE
UNIVERSITAS GADJAH MADA
2026

FINAL PROJECT

FINAL PROJECT TITLE (ENGLISH)

Submitted in partial fulfillment of the requirements for the Applied Bachelor's Degree
in Engineering in the Study Program Electrical Engineering Technology Department
of Electrical Engineering and Informatics Vocational College Universitas Gadjah
Mada

By:

FULL NAME

XX/XXXXXX/SV/XXXXX

**APPLIED BACHELOR'S DEGREE PROGRAM
ELECTRICAL ENGINEERING TECHNOLOGY
DEPARTMENT OF ELECTRICAL ENGINEERING AND INFORMATICS
VOCATIONAL COLLEGE
UNIVERSITAS GADJAH MADA
2026**

**APPROVAL PAGE
FINAL PROJECT**

Final Project Title (English)

Prepared by:

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Study Program Electrical Engineering Technology

**Has been defended and accepted by the Examination Committee
as one of the requirements for obtaining the Applied Bachelor's Degree**

Examination Date: August 14, 2026

Chair Examiner

Name of Chair Examiner

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Name of Program Head

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Date of Approval: _____

Acknowledged by,

Head of Department of Electrical Engineering and Informatics

Name of Department Head

NIP. XXXXXXXXXXXXXXXXXXXXX

DECLARATION LETTER

I, the undersigned below:

Name : Full Name
Student ID : XX/XXXXXX/SV/XXXXX
Year of Enrollment : XXXX
Study Program : Electrical Engineering Technology
Faculty : Vocational College

Hereby declare that this Final Project scientific document does not contain any part of other scientific works that have been submitted for obtaining an academic degree at any higher education institution, and also does not contain any work or opinion that has been written or published by other persons/institutions, except those that are cited in writing in this document and listed in full in the bibliography.

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Yogyakarta, August 14, 2026

The declarant,

Full Name
XX/XXXXXX/SV/XXXXX

MOTTO PAGE

"quotation"
(Quoted figure)

PREFACE

All praise and gratitude to God Almighty for His blessings and grace, this Final Project has been completed in order to fulfill part of the requirements for obtaining an Applied Bachelor's Degree.

This Final Project could not have been completed without the assistance and cooperation of various parties. In this regard, the author would like to express gratitude to the following:

1. Acknowledgment 1
2. Acknowledgment 2
3. Acknowledgment 3
4. Acknowledgment 4
5. Acknowledgment 5

Finally, may all the assistance provided by all parties become beneficial deeds and receive good rewards both now and in the future. Hopefully, this Final Project will become useful information for readers or other parties who may need it.

Yogyakarta, August 14, 2026

Full Name
XX/XXXXXX/SV/XXXXX

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LIST OF ABBREVIATIONS

Notation	Meaning
PID	<i>Proportional, Integral, Derivative</i>
K_p	<i>Proportional gain</i>
K_i	<i>Integral gain</i>
K_d	<i>Derivative gain</i>

INTISARI

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Kata kunci: keyword 1, keyword 2, keyword 3

ABSTRACT

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Key words: keyword 1, keyword 2, keyword 3

CHAPTER I

INTRODUCTION

1.1 Background

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1.2 Problem Formulation

The following are several points that have been identified and formulated as the main problems that will be the focus of this research [1]:

1. Problem formulation 1
2. Problem formulation 2
3. Problem formulation 3

1.3 Scope of Research

In order to keep the research focused on the established scope [2], the following limitations need to be considered in the development of this research:

1. Scope limitation 1
2. Scope limitation 2
3. Scope limitation 3

1.4 Research Objectives

The following are several research objectives that have been established to guide the course of this research:

1. Research objective 1

2. Research objective 2
3. Research objective 3

1.5 Research Benefits

The benefits obtained from this research are as follows:

1. Research benefit 1
2. Research benefit 2
3. Research benefit 3

1.6 Writing Systematics

The final project report is written with the following writing systematics.

1. Chapter I Introduction: explains the background, problem formulation, objectives, scope of research, and writing systematics of the applied bachelor's final project report.
2. Chapter II Literature Review: describes the studies conducted by previous researchers related to the topic investigated in the applied bachelor's final project, and discusses theories relevant to the problems being studied. This chapter contains a literature review obtained from various sources related to the problems being studied.
3. Chapter III Research Methodology: describes the steps and design carried out in the research, as well as the methods used for the approach in this research.
4. Chapter IV Results and Discussion: describes the results obtained from the applied bachelor's final project and provides in-depth discussion related to those results. This chapter also contains the interpretation of data obtained from the research.
5. Chapter V Conclusion and Recommendations: describes the conclusions obtained from the applied bachelor's final project and provides recommendations for future research.
6. References: describes the sources used in the applied bachelor's final project report.

Overall, the writing systematics in the applied bachelor's final project report is the arrangement or structure of the final project report that describes the parts that must be included in the report, which consists of Introduction, Literature Review,

Research Methodology, Results and Discussion, Conclusion and Recommendations, and References. Good writing systematics will make the applied bachelor's final project report easier to read and understand.

CHAPTER II

LITERATURE REVIEW

2.1 Literature Review

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Table 2.1 Example Table 1

Author	Title	System Modeling Method
Author 1	Title 1	System Modeling 1
Author 2	Title 2	System Modeling 2
Author 3	Title 3	System Modeling 3
Author 4	Title 4	System Modeling 4
Author 5	Title 5	System Modeling 5
Author 6	Title 6	System Modeling 6

Table 2.2 Example Table 2

No.	Name	Function
1	<i>Detail object name to be discussed 1</i>	Detail object function description 1
2	<i>Detail object name to be discussed 2</i>	Detail object function description 2
3	<i>Detail object name to be discussed 3</i>	Detail object function description 3
4	<i>Detail object name to be discussed 4</i>	Detail object function description 4

2.2 Theoretical Foundation

2.2.1 Theoretical Foundation Topic 1

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2.2.2 Theoretical Foundation Topic 2

A. Sub-topic Explanation 2

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$$\begin{aligned} Tf = G(s) &= \frac{\text{output}}{\text{input}} \\ Tf &= \frac{b_0 s^m + b_1 s^{m-1} + \dots + b_{m-1} s + b_m}{a_0 s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n} \end{aligned} \tag{2.1}$$

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B. Sub-topic Explanation 2

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1. Explanation point 1
2. Explanation point 2
3. Explanation point 3
4. Explanation point 4

C. Sub-topic Explanation 3

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$$linear(x; a, b) = \begin{cases} 0; & x \leq a \\ (x - a)/(b - a); & a \leq x \leq b \\ 1; & x \geq b \end{cases} \quad (2.2)$$

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CHAPTER III

RESEARCH METHODOLOGY

The methodology applied in the research is the main subject of this chapter. The research planning workflow, data collection and analysis, and research results will be explained in detail. In addition, the explanations in this chapter also cover other components such as research approaches, data collection techniques, types of data collected, analysis methods, and supporting instruments used.

3.1 Research Stages

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3.2 Research Stage Explanation 1

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3.2.1 Sub-explanation of Research Stage 1

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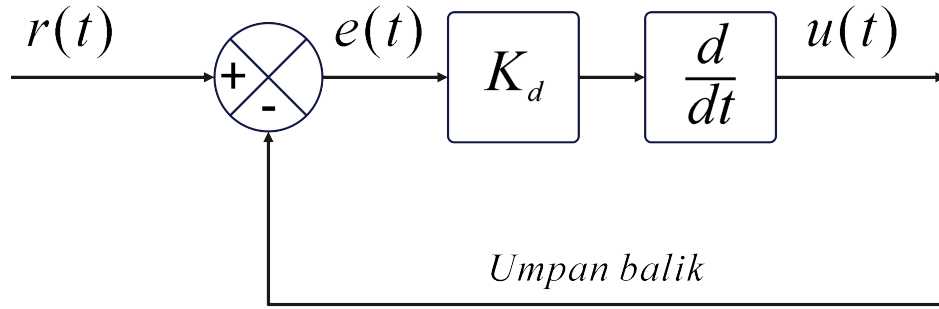


Figure 3.1 Example figure 1

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3.2.2 Sub-explanation of Research Stage 1

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3.3 Research Stage Explanation 2

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3.3.1 Sub-explanation of Research Stage 2

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Table 3.1 Example table 3

	$\dot{e}(k)$							
$e(k)$		NB	NM	NS	ZO	PS	PM	PB
	NB	S	S	S	S	S	S	S
	NM	MS	MS	S	S	S	MS	MS
	NS	M	MS	MS	S	MS	MS	M
	ZO	B	M	MS	MS	MS	M	B
	PS	M	MS	MS	S	MS	MS	M
	PM	MS	MS	S	S	S	MS	MS
	PB	S	S	S	S	S	S	S

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 Discussion Results Explanation 1

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4.1.1 Data Collection

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Table 4.1 Example table 4

Variation	K_p	K_i	K_d
PID1	1	1	1
PID2	3	1	1
PID3	4	7	8
PID4	4	5	9

A. Data Variation 1

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B. Data Variation 2

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C. Data Variation 3

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4.1.2 Transfer Function Estimation

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Table 4.2 Example table 5

		<i>Validation data</i>					
		Data-1	Data-2	Data-3	Data-4	Data-5	Data 6
<i>Transfer function</i>	TF_1	83,13	77,73	80,74	94,22	85,88	83,76
	TF_2	79,61	83,34	76,93	84,3	82,39	77,31
	TF_3	81,88	72,91	82,17	95,17	84,9	81,9
	TF_4	82,6	73,58	81,63	96,21	85,44	84,23
	TF_5	83,13	77,33	80,95	94,59	85,89	83,87
	TF_6	83,56	73,29	81,42	96,17	85,4	84,33

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Table 4.3 Example table 6

Characteristic	Value
<i>Rise time</i> (s)	1,0580
<i>Settling time</i> (s)	1,3380
<i>Overshoot</i> (%)	0,0910
<i>Peak</i>	1,0910
<i>Peak time</i> (s)	1,0840

4.2 Summary of Test Results

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Table 4.4 Example table 7

Characteristic	Performance comparison			
	PID	MF3	MF5	MF7
<i>Rise time</i> (s)	1,0580	1,0740	1,0620	1,0450
<i>Settling time</i> (s)	1,3380	1,3410	1,3350	1,3290
<i>Overshoot</i> (%)	0,0910	0,0440	0,0593	0,0121
<i>Peak</i>	1,0910	1,0440	1,0593	1,0121
<i>Peak time</i> (s)	1,0840	1,3560	1,2710	1,0520

CHAPTER V

CONCLUSION

5.1 Conclusions

Based on the analysis and discussion of the test results data, several conclusions were obtained, namely:

1. Conclusion 1
2. Conclusion 2
3. Conclusion 3

5.2 Recommendations

Based on the research results conducted, several weaknesses were found in the implementation, thus further development efforts are needed to improve the existing system. The following are recommendations that can be used as guidelines for future research development:

1. Recommendation 1
2. Recommendation 2
3. Recommendation 3

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APPENDIX A

APPENDIX B